Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/11/2005
PI Name:	Pickering, Karen D Ph.D.		
Project Title:	Minimization of Ultrafiltration Membrane Fouling through the Optimization of Membrane Surface/Surfactant Interactions		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-2688
Organization Name:	NASA - Johnson Space Center		
PI Address 1:	EC3		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	01/01/2002	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Original end date was 10/1/2005; changed to 9/30/2004 when project was cancelled in FY2004 (jp 2/9/09)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The project "Minimization of Ultrafiltration Membrane Fouling through the Optimization of Membrane Surface/Surfactant Interactions" is an effort to improve the operational performance of the first stage of the ultrafiltration/reverse osmosis water processor used in the Lunar-Mars Life Support Test Project Phase II test. In order to minimize the fouling of membranes used in the first stage of this system, the fouling of ultrafiltration membranes by surfactants found in personal care products will be evaluated. The surfactants and membrane materials will be studied to determine if an optimum combination of materials can be found to minimize fouling and maximize contaminant removal. The relationship between micellar properties and ultrafiltration performance will be examined to determine if a generic relationship can be developed to predict the suitability of surfactant mixtures for treatment with ultrafiltration. Ultimately, this information will be used to reduce the equivalent system mass of an ultrafiltration/reverse osmosis system. A variety of surfactants and membrane materials will be studied to determine if an optimum combination of materials can be found to minimize fouling and maximize contaminant removal. The project "Minimization of Ultrafiltration Membrane Fouling through the Optimization of Membrane Surface/Surfactant Interactions" is an effort to improve the operational performance of the first stage of the ultrafiltration/reverse osmosis water processor. This information will be used to reduce the equivalent system mass of an ultrafiltration/reverse osmosis system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are terrestrial situations with significant surfactant concentrations in their wastewater that are suitable applications for membrane filtration. Wastewater from residential areas has a significant proportion of surfactants. In addition, wastewater from industrial applications such as textile processing will have high concentrations of surfactants. Information from this effort has the potential to be used to develop water reuse systems appropriate for both spacecraft and terrestrial applications.
Task Progress:	This project was cancelled at the start of FY04.
Bibliography Type:	Description: (Last Updated: 09/21/2006)